



Immunodiagnostic potential of *Wuchereria bancrofti* L1 antigen-based filarial immunoglobulin G4 detection assay

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Background: After mass drug administration to eliminate human lymphatic filariasis, there is a need for surveillance to detect the measurable endpoint of the program.

Methods: An immunodominant seroreactive clone, *WbL1*, was identified through immunoscreening of a *Wuchereria bancrofti* L3 complementary DNA expression library. Recombinant *WbL1* (r*WbL1*) was analysed with sera from *W. bancrofti* patients. Diagnostic evaluation was carried out by developing an enzyme-linked immunosorbent assay (ELISA) to detect the filarial-specific antibodies in various categories of filarial sera samples against recombinant *WbL1* (r*WbL1*) protein.

Results: Performance parameters of the test in terms of immunoglobulin G (IgG) and IgG4 detection displayed significant sensitivity and specificity values up to 77% and 100%, respectively. Our results showed filarial antibodies against r*WbL1* to be highly reactive with microfilaremic and clinical filarial sera samples compared with the endemic and non-endemic control sera samples. Reasonably satisfactory performance of the test was also confirmed from the multicentric evaluation of an anti-*WbL1* IgG4 detection ELISA. This test was found to be minimally reactive with other nematode parasites and protozoan infections.

Conclusions: The anti-*WbL1* IgG4 detection test can be considered as a field test for initial screening and epidemiological monitoring of filarial infections in filariasis-endemic areas.

Keywords: *Brugia malayi*, diagnostic assay, epidemiological indicator, immunoscreening, lymphatic filariasis, *Wuchereria bancrofti*

Introduction

Lymphatic filariasis (LF), a mosquito-borne disorder in humans, is caused by tissue-dwelling parasites. The Global Program to Eliminate Lymphatic Filariasis (GPELF), which was initiated in 2000, has delivered anti-filarial drugs to more than 6.2 billion individuals in affected areas.¹ In this programme, mass drug administration (MDA) was implemented in 73 countries where LF is endemic. Of these, 18 have successfully completed their MDA program, 22 have provided MDA and 33 are in the process of implementing their MDA program.¹ The success of the GPELF program has emphasized the need for tools that can be used to monitor progress and to look for measurable endpoints (e.g. when to stop mass treatment) as well as to conduct surveillance to detect any potential need for resumption of MDA. Measurement of microfilaremia, the recognized gold standard

for demonstrating the impact of community-wide intervention, is not an optimal monitoring and surveillance tool. It has been criticized because of its need for nocturnal blood collection, which is practically difficult at the field level, and also because it is a relatively less sensitive test for detection of infection.² One of the major reasons for this is that microfilariae are not usually seen in the peripheral circulation in acute, chronic and occult filariasis.

Transmission assessment surveys (TASs), which are part of post-MDA surveillance studies, mainly rely on the detection of adult-specific circulating filarial antigen.³ The antigen detection-based diagnostic kits manage to detect patent infection but fail to determine the course of infection in terms of antibody levels, which is the main drawback of this assay. Moreover, both microfilaremia and antigenemia develop from months to years after exposure, reducing their utility for detection of low levels of infection or recurrence of transmission.^{4,5} In this context,

detection of specific antibodies against third-stage larva (L3) might be better suited for this purpose, given that these antibodies may be detectable significantly earlier (months at least) than the conventional detection of circulating antigen, particularly for children 6–7 y old, those who are targeted as the sentinel population for TAs and surveillance.^{3,6,7}

Antigen- or antibody-based immunodiagnostic tests with various degrees of sensitivity and specificity have been reported.^{2,8,9} Even though antigen detection assays are reasonably sensitive and specific for identification of individuals with active infection, development of a rapid antibody test can be used for the diagnosis of clinical cases and for early detection of infection in children, travellers and expatriates. Such an assay will also be highly useful for primary surveillance in areas of unknown endemicity and for monitoring chemotherapeutic programs.⁵

In the present study, the complementary DNA (cDNA) library of *Wuchereria bancrofti* L3 stage was screened with pooled sera of microfilaremic and non-endemic normal individuals. This approach identified a novel gene that showed significant similarity to an immunogenic protein from filarial parasites. This study describes the cloning, characterization and seroreactivity of this novel protein from *W. bancrofti* and explores its diagnostic potential.

Materials and methods

Ethical approval and collection of human sera samples

This study was approved by the Human Ethics Committee of the Mahatma Gandhi Institute of Medical Sciences, Sevagram, India. Night blood smear analysis was carried out to detect adults residing in filarial-endemic regions of Sevagram and surrounding villages located in central India with active filarial infection. After obtaining proper informed written consent in both English and the native language from all the adults participating in this study, 10 ml of blood was collected in blood collection tubes through intravenous vein puncture; serum samples were isolated and kept frozen at -80°C for further use.

The samples were classified into three major groups according to the category of subjects: microfilaria (Mf) positive—individuals who had active filarial infection and were positive for circulating Mf in night blood smear analysis; endemic normal (EN)—individuals who are negative on night blood smear analysis and do not show any clinical symptoms of the disease; clinical filarial (CL)—individuals with disease manifestations such as lymphedema, elephantiasis or hydrocele; and non-endemic normal (NEN)—individuals residing in non-endemic areas for filarial infection.

Immunoscreening of *W. bancrofti* L3 cDNA library

A lambda Zap L3 cDNA expression library of *W. bancrofti* (kindly provided by Dr Steven Williams, Smith College, Northampton, MA, USA) was immunoscreened as described previously with a few modifications.¹⁰ Briefly, the *W. bancrofti* L3 cDNA library was plated using XL1-Blue MRF strain and grown at 42°C for 3 h. It was later overlaid with 1 mM isopropyl β -D-thiogalactopyranoside (IPTG)-impregnated nitrocellulose membranes (Millipore, Mumbai, India) and grown at 37°C for an additional 4 h. The filters were then removed, blocked with 5% skim milk powder in 15 mM Tris-buffered saline (TBS, pH 7.5). Filters were then probed with optimally diluted (1:100) pooled

Mf and NEN sera that were earlier absorbed with *Escherichia coli* phage lysate to remove anti-*E. coli* antibodies to reduce cross-reactivity. Secondary antibody, alkaline phosphatase-conjugated anti-human IgG (Sigma-Aldrich, Mumbai, India), was used at a dilution of 1:5000. The antigen antibody complexes were visualized using nitro blue tetrazolium and 5-bromo-4-chloro-3-indolyl phosphate. Common reactive clones were omitted and only the clones reacting specifically with Mf sera were selected. Positive clones were purified by secondary and tertiary levels of immunoscreening at a lower density of plaques.

Polymerase chain reaction (PCR) and sequence analysis of the immune-positive plaques from the L3 cDNA library

Positive plaques from the L3 cDNA library were transferred into 50 μl of SM buffer (0.1 M NaCl, 50mM Tris-HCl [pH 7.5], 8 mM MgCl_2 and 0.01% gelatin). To amplify the random inserts, PCR was performed in 5 μl aliquots of plaque with T3 and T7 vector primers. PCR products were then purified using the QIA quick purification kit (Qiagen, Valencia, CA, USA) and DNA sequencing was performed. Sequences were examined for similarity to the already available sequences in the GenBank non-redundant nucleic acid database using BLAST analysis (National Centre for Biotechnology Information, Bethesda, MD, USA; <http://www.ncbi.nlm.nih.gov>).

Expression and purification of recombinant WbL1

The sequence of the *WbL1* gene has been deposited in the GenBank database under accession number EF455880. The open reading frame (ORF) of the selected clone *WbL1* was cloned in T 7 expression vector, pRSET-B (Thermo Fisher Scientific, Mumbai, India). Forward (5'-CCTTAAGGATCCGTCACCTTCATCACTCAAT-3') and reverse (5'-AACCAAGAATTCCTATTATTACTTTTGTGCG -3') PCR primers were designed with the addition of *Bam*HI and *Eco*RI restriction sites, respectively. Cloned *WbL1*_pRSET-B plasmid was sequenced and transformed into osmotically (salt) inducible *E. coli* GJ1158 host with salt (NaCl) as inducer.¹¹ Histidine-tagged recombinant protein was purified by immobilized metal affinity of a Ni^{2+} -charged chromatography column (Invitrogen Bioservices). The expression profile and purity of the recombinant protein was subsequently determined by sodium dodecyl sulphate–polyacrylamide gel electrophoresis (SDS-PAGE) analysis and confirmed by western blot analysis using penta-His antibody (Qiagen). The protein concentration was determined using a BCA protein assay kit (Thermo Fisher Scientific).

Immunoblot analysis of recombinant WbL1 (rWbL1) antigen

Immunoblot analysis of rWbL1 antigen was carried out using optimally (1:100) diluted sera of Mf, EN, CL, NEN and anti-His tag horseradish peroxidase (HRP)-conjugated antibody.

Determination of rWbL1-specific antibody levels in the sera from filarial endemic regions

Antigen-specific IgG and IgG4 antibodies were determined by an indirect ELISA. The wells of microtitre plates (Thermo Fisher

Scientific) were coated with 50 ng/well of rWbL1 in carbonate bicarbonate buffer (pH 9.6) overnight at 4°C. Levels of antigen-specific IgG and IgG4 antibodies in diluted sera samples (1:100 diluted in phosphate-buffered saline [PBS]) were determined using optimally diluted HRP-conjugated goat anti-human IgG (Sigma-Aldrich) or biotinylated mouse monoclonal anti-human IgG4 antibodies (1:25 000 diluted in PBS; Sigma-Aldrich). Avidine peroxidase conjugate (1:30 000 diluted in PBS; Sigma-Aldrich) was used in the assay where biotinylated antibodies are used as a secondary antibody. The colour reaction was developed using 3,3',5,5'-tetramethylbenzidine/H₂O₂ substrate (Sigma-Aldrich)

and the reaction was stopped by the addition of 2N H₂SO₄. Absorbance was measured at 450 nm using an ELISA reader (Biotek, Mumbai, India).

Stability studies on rWbL1

A short-term stability study was performed as described.¹² Immunomodules were sensitized with rWbL1, blocked and incubated at different temperatures (4°C, 37°C and 45°C) and checked for stability. The antigens were tested against dilutions of high titres of serum samples and test results were compared

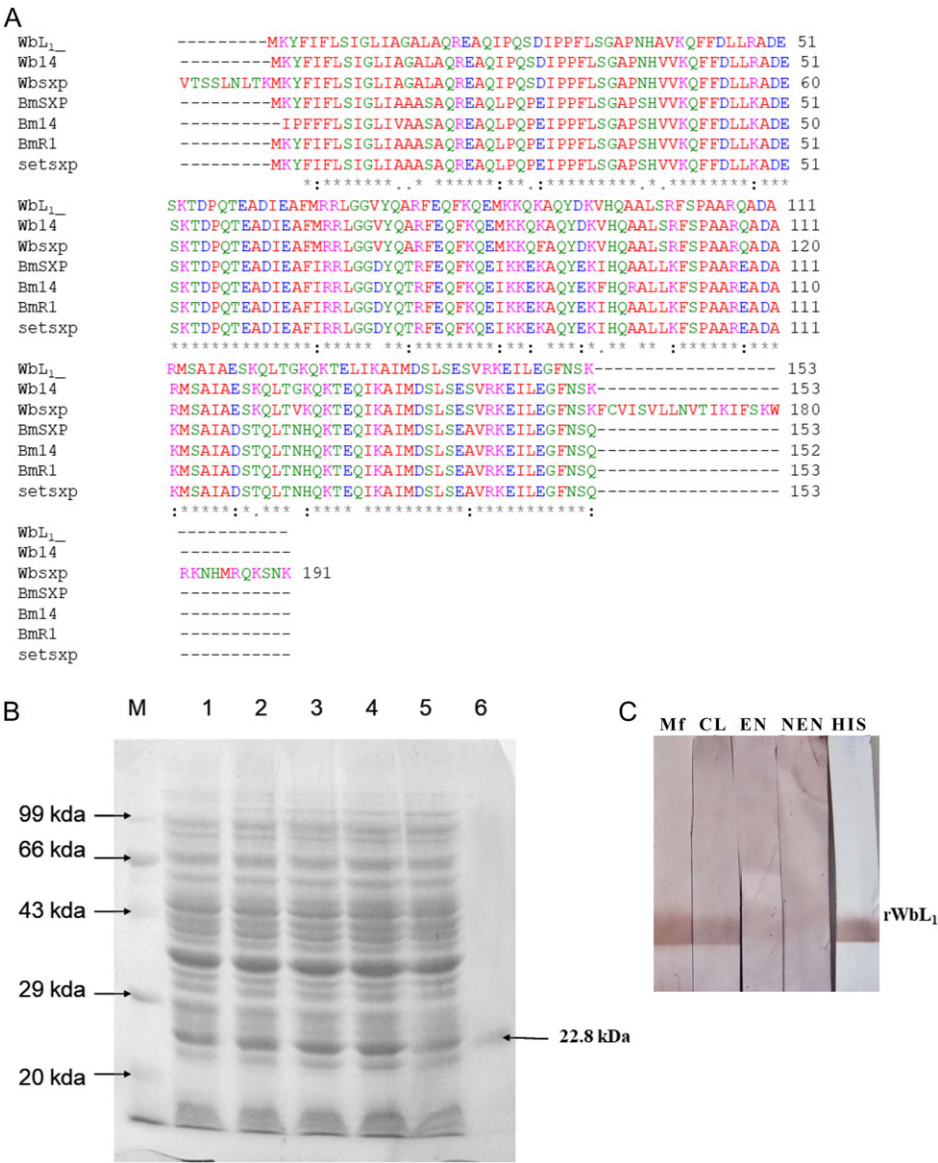


Figure 1. Expression of the recombinant WbL1 protein. (A) Alignment (clustal) of the amino acid sequences of members of the SXP/RAL2 family of proteins. *Wuchereria bancrofti* (WbL1, accession no. EF455880), *W. bancrofti* (Wb14 accession no. AF63940), *W. bancrofti* (Wb-SXP-1, accession no. AF098861), *Brugia malayi* (Bm SXP-1, accession no. M98813), *B. malayi* (Bm14 accession no. S76367), *B. malayi* (BmR1 accession no. AF225296) and *Schefflera digitata* (SdSXP-1 accession no. AY928311). (B) SDS-PAGE (14%) analysis of the expressed and purified recombinant WbL1 protein (lanes: M, molecular weight markers 1, 2, 3 and 4; cell culture lysates after 1, 2, 3 and 4 h of induction; 5, Uninduced culture; 6, rWbL1 purified protein). (C) Immunostaining of rWbL1 antigen probed with the pooled sera of microfilaremic individuals (Mf), clinical filarial cases (CL), endemic normal individuals (EN), non-endemic normal individuals (NEN) and anti-His tag antibody (HIS).

and analysed accordingly. For the long-term stability studies, assay plates were stored for a period of 1, 3 and 6 months.

Statistical analysis

The statistical analysis was carried out using SPSS 21.0 (IBM, Armonk, NY, USA) and suitable parametric or non-parametric tests were applied based on normality assumptions of the data. p-Values <0.05 were considered statistically significant. Sensitivity, specificity, positive predictive value (PPV) and negative predictive value (NPV) were also calculated to assess the diagnostic attributes of the antibody detection assay by rWbL1 antigen against night blood smear analysis as the gold standard.

Results

Immunoscreening of *W. bancrofti* L3 cDNA library and expression of rWbL1 protein

More than 50 000 phages were screened through 10 lambda phage plates (0.5×10⁴ plaques/140 mm petri plate). In the preliminary screening, about 50 plaques were identified to be more reactive with Mf sera compared with NEN sera. Later, secondary and tertiary screening of positive plaques identified seven strongly positive plaques reacting with the Mf sera samples. Blast analysis showed that one of the seven identified clones belongs to the SXP/RAL-2 family of proteins and may have diagnostic potential (Figure 1A). This clone was selected for exploration of its serological diagnostic potential. The gene was named *WbL1* and it was sequenced and submitted to the National Center for Biotechnology Information (NCBI) database (accession no. EF455880.1). The *WbL1* gene was approximately 626 bp with an ORF of 462 bp. Utilizing the putative start codon

as a reference point, the cDNA contained a 32-bp 5' untranslated region. The 132 bp that made up the 3' untranslated region contained a polyadenylation signal at 574–579 bp. The expressed recombinant protein was approximately 22.8 kDa in size based on the SDS-PAGE analysis (Figure 1B). Recombinant protein was confirmed with the western blot analysis using anti-His tag antibodies (Figure 1C). The immunoreactivity of the recombinant protein with human sera using the immunoblot analysis showed that rWbL1 antigen was specifically reactive with the sera of Mf and CL individuals (Figure 1C), whereas no significant reactivity was observed with EN and NEN sera samples (Figure 1C).

Diagnostic efficacy of rWbL1 protein for filarial infection by indirect IgG antibody ELISA

A total of 150 sera samples belonging to distinct categories of filariasis (Mf, CL, EN and NEN) were screened for anti-WbL1 IgG antibodies using indirect ELISA. The results showed significant (p<0.05) levels of anti-WbL1 IgG antibodies present in the sera of Mf and CL individuals compared with EN and NEN individuals (Figure 2). Considering the mean+3 standard deviation (SD) absorbance value of the NEN group as the lower limit of positivity, 55 of 80 (68.8%) Mf sera and 7 of 20 (35%) CL sera were positive for filarial antibodies (Table 1). None of the 20 EN and 30 NEN sera were found to be positive for filarial antibodies (Table 1). The sensitivity and specificity of the ELISA for detection of the anti-WbL1 IgG antibodies were 68.8% and 100%, respectively, and the PPV and NPV of the assay were 100% and 66.6%, respectively.

Detection of the levels of anti-WbL1 IgG4 antibodies in the filarial sera samples

A total of 235 human sera samples belonging to the four distinct groups (Mf, CL, EN and NEN) were analysed for anti-rWbL1 IgG4 antibodies using indirect ELISA. Considering the mean+3 SD absorbance value of the NEN group as the lower limit of positivity, 77 of 100 (77%) Mf sera, 15 of 45 (33%) CL sera and 1 of 45 (2.2%) each EN and NEN sera samples were positive for anti-WbL1 IgG4 antibodies (Table 2). Results showed the presence of a significant (p<0.05) amount of anti-rWbL1 IgG4 antibodies in the sera of Mf individuals compared with sera from CL, EN and

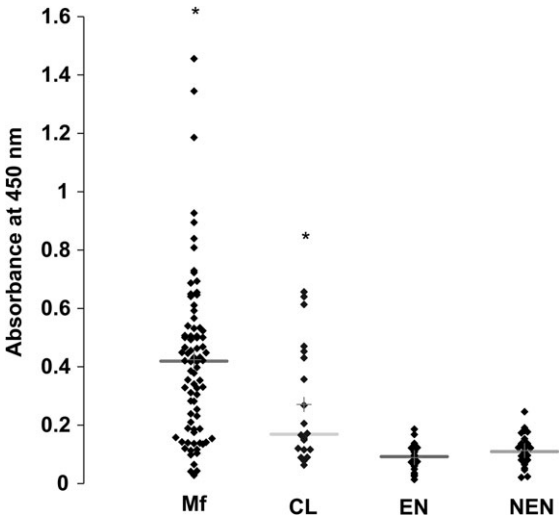


Figure 2. Filarial IgG antibody levels against rWbL1 in distinct categories of filarial sera samples. Total IgG antibody levels against rWbL1 in serum were analysed using indirect ELISA. Each data point represents absorbance values of individual groups of 80 Mf, 20 CL, 20 EN and 30 NEN sera samples. The solid line indicates the geometric mean value in each group. *p≤0.05 compared with all other groups as analysed by Kruskal–Wallis test followed by Bonferroni correction for multiple groups.

Table 1. Detection of anti-WbL1 IgG antibodies in different groups of filarial sera

Sera sample categories	Samples screened, n	Positive sera samples for anti-WbL1 IgG antibodies*, n (%)
Microfilaremia	80	55 (68.8)
Clinical filariasis	20	7 (35)
Endemic normal	20	0 (0)
Non-endemic normal	30	0 (0)

*Sera samples showing absorbance values ≥mean+3 SD of NEN sera were considered positive.

Table 2. Detection of anti-WbL1 IgG4 antibodies in different groups of filarial sera

Sera sample categories	Samples screened, n	Positive sera samples for anti-WbL1 IgG4 antibodies*, n (%)
Microfilaremia	100	77 (77)
Clinical filariasis	45	15 (33)
Endemic normal	45	1 (2.2)
Non-endemic normal	45	1 (2.2)

*Sera samples showing absorbance values $\geq$ mean+3 SD of NEN sera were considered positive.

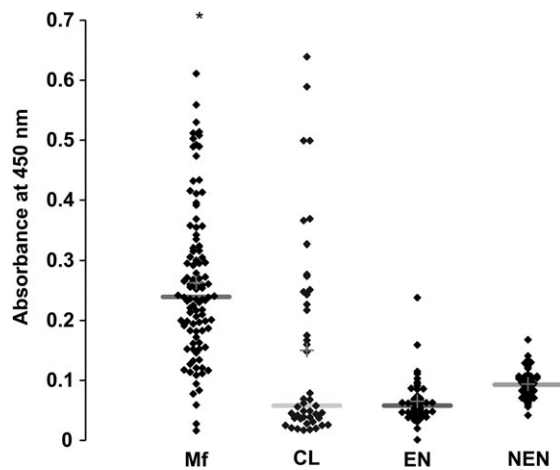


Figure 3. Filarial IgG4 antibody levels against rWbL1 in distinct groups of bancroftian filariasis. IgG4 antibody levels against rWbL1 in serum were analysed using indirect ELISA. Each data point represents absorbance values of individual groups of 100 Mf, 45 CL, 45 EN and 45 NEN sera samples. The solid line indicates the geometric mean value in each group. * $p \leq 0.05$ compared with all other groups as analysed by Kruskal–Wallis test followed by Bonferroni correction for multiple groups.

NEN individuals (Figure 3). The mean absorbance value of rWbL1-specific IgG4 antibody levels in Mf sera ($A_{450}=0.262$) was significantly higher than in EN ($A_{450}=0.065$) and NEN sera ($A_{450}=0.093$). The detection of anti-WbL1 IgG4 antibodies increased the sensitivity and specificity of the test to 77% and 97.8%, respectively. The PPV and NPV for the test were observed to be 97.4% and 79.6%, respectively.

Multicentric validation of the rWbL1 antigen-based IgG4 antibody assay

A total of 409 sera samples of distinct categories from the endemic areas of Chennai (Tamil Nadu, India), Institute of Vector Control and Zoonoses (Hosur, Tamil Nadu, India) and Regional Medical Research Centre (Dibrugarh, Assam, India) were used to evaluate the diagnostic potential in bancroftian filariasis based on the detection of filarial IgG4 antibodies. A significant ($p < 0.05$) level of rWbL1-specific filarial IgG4 antibodies were observed in the Mf compared with the CL, EN and NEN samples (Figure 4).

Table 3. Multicentric validation of the rWbL1 antigen-based IgG4 antibody assay

Sera sample categories	Total samples screened, n	Positive sera samples for anti-WbL1 IgG4 antibodies*, n (%)
Microfilaremia	146	125 (85.61)
Clinical filariasis	74	37 (50)
Endemic normal	79	10 (12.65)
Non-endemic normal	110	1 (0.9)
<i>Paragonimus</i> , Kala-azar and Japanese encephalitis	18	2 (11.11)

*Sera samples showing absorbance values $\geq$ mean+3 SD of NEN sera were considered positive.

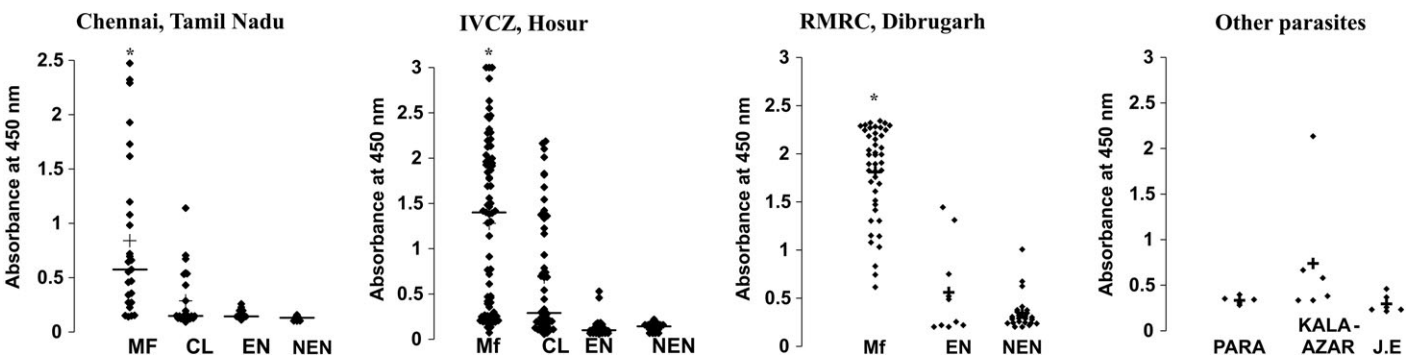


Figure 4. Scattergram showing filarial IgG4 antibodies detected in distinct groups of human sera analysed at Chennai (Tamil Nadu, India), Institute of Vector Control and Zoonoses (IVCZ, Hosur, Tamil Nadu, India) and Regional Medical Research Centre (RMRC, Dibrugarh, Assam, India) against rWbL1 antigen by indirect ELISA. Antibody levels against serum samples from other parasitic infections are also analysed. Each data point represents absorbance values of individual serum samples and the solid line indicates the geometric mean value in each group (PARA: *Paragonimus*; JE: Japanese encephalitis). * $p \leq 0.05$ compared with all other groups as analysed by Kruskal–Wallis test followed by Bonferroni correction for multiple groups.

Considering the mean optical density value plus 3 SD of NEN to define the upper limit of the normal reference, overall 85.61% of Mf sera and 37% of CL sera were found to be positive for the presence of anti-WbL1 IgG4 antibodies, while around 12.65% of EN sera and only 1% of NEN sera samples were observed to be positive for anti-WbL1 IgG4 antibodies (Table 3). This multicentric evaluation of the rWbL1 antigen-based IgG4 detection diagnostic assay displayed a maximum sensitivity and specificity of 93% and 98%, respectively; the maximum recorded PPV value was 96% and NPV was 94% (Table 4). The overall sensitivity and specificity of the test were 85% and 94%, respectively, and both the PPV and NPV were 90% (Table 4).

Stability studies

The rWbL1 antigen was tested for its antigenic stability by exposing it to different temperatures for varying periods. Short-term stability studies revealed that it is stable at 4°C. It was shown to be stable under long-term storage at 4°C for a period of 3 months (Figure 5).

Discussion

To ensure the success of GPELF by year 2020, diagnostic tools will play a significant role in mapping of endemicity, monitoring the effectiveness of MDA and maintaining post-elimination surveillance. Hence there is a growing need for a serodiagnostic test that is simple, specific and rapid for use in field-level studies

for detecting both bancroftian as well as brugian filarial infection. In the present study, WbL1 protein showed a significant resemblance with the SXP/RAL-2 family of proteins, which are known for their diagnostic importance.^{13,14} The nucleotide analysis of a cloned WbL1 gene showed high AT content (62%) which resembles the characteristic AT-rich pattern of various filarial parasite genes reported earlier.¹⁵ During the immunoscreening of expressed WbL1 protein, it reacted strongly with pooled Mf sera but not with NEN sera. Hence, considering the diagnostic potential of rWbL1, an attempt was made to evaluate the newly developed ELISA test for the detection of anti-WbL1 IgG and IgG4 antibodies from the sera of distinct groups of filarial patients. The sensitivity (maximum up to 68.8%) and specificity (almost 100%) of IgG ELISA test was found to be on a par with the previously reported ELISA utilizing the SXP/RAL-2 family of proteins.^{8,16} The seroreactivity of rWbL1 was significantly higher in the Mf and CL groups compared with those in the EN and NEN groups, as indicated by the mean absorbance values in the ELISA. These results indicate that the peptide epitope being expressed by the gene sequence of WbL1 is serospecific for antibodies generated by human subjects infected with *W. bancrofti*. Possibly being a larval protein, the chance of detection of an antibody against this makes it a quite sensitive, early response test as well. Earlier studies have provided evidence in favour of IgG-based detection of filarial infection.^{17,18}

Asymptomatic filarial infections are characterized by high plasma levels of IgG4 antibodies.¹⁹ Thus an assay based on the IgG4 antibody isotype might be promising as an immunodiagnostic test,

Table 4. Performance characteristics of the multicentric validation of the rWbL1 antigen-based IgG4 antibody assay

Centre name	Mf positive		Mf negative		Total (a+b)	Total (c+d)	Total (a+c)	Total (b+d)	Sensitivity [a/(a+b)] *100	Specificity [d/(c+d)]*100	PPV [a/(a+c)]*100	NPV [d/(b+d)]*100
	ELISA positive (a)	ELISA negative (b)	ELISA positive (c)	ELISA negative (d)								
Chennai	23	4	6	31	27	37	29	35	85.18	83.78	79.31	88.57
IVCZ	61	14	2	118	75	120	63	132	81.33	98.33	96.82	89.39
RMRC	41	3	5	45	44	50	46	48	93.18	90	89.13	93.75
Total	125	21	13	194	146	207	138	215	85.61	93.71	90.57	90.23

IVCZ: Institute of Vector Control and Zoonoses (Hosur, Tamil Nadu, India); RMRC: Regional Medical Research Centre (Dibrugarh, Assam, India).

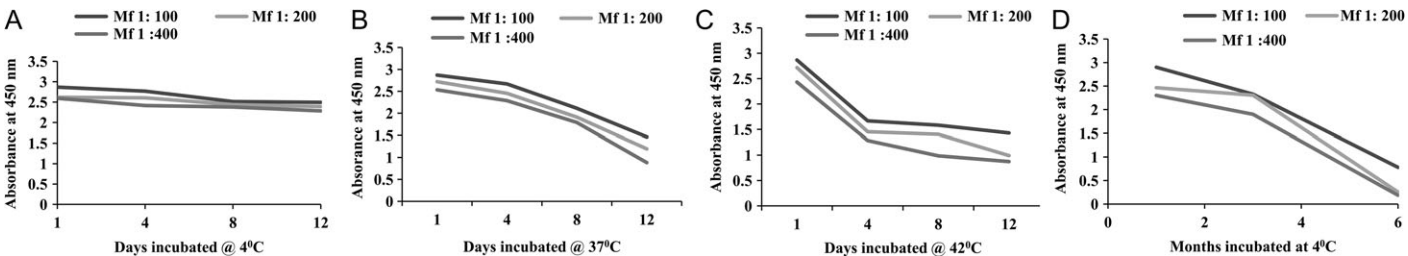


Figure 5. Stability studies for the IgG4 rWbL1 ELISA filarial diagnostic test. The antigen activity of rWbL1 was measured based on its reactivity with pooled microfilaremic serum and expressed as an absorbance value (at 450nm) in an indirect microtitre plate. Plates incubated at (A) 4°C, (B) 37°C, (C) 45°C and (D) 4°C up to 6 months.

hence we evaluated rWbL1-specific IgG4 antibodies in different groups of bancroftian filariasis. The performance characteristics (i.e., sensitivity and specificity) of the rWbL1-based ELISA test increased with detection of IgG4 antibodies compared with when used for the detection of IgG antibodies. Based on these results, we evaluated our test through a double-blind, multicentric evaluation carried out at three different centres. The rWbL1 antigen-based diagnostic test showed high performance characteristics, with sensitivity and specificity of about 85% and 94%, respectively. The efficacy of the test was observed to be significantly high, as none of the sera from non-endemic areas showed positive results. Moreover, initial data from a small-scale study showed that this IgG4 antibody detection test was minimally reactive with other nematode parasites and protozoan infections, as observed by other investigators,²⁰ thus once again confirming the specificity of the test. The stability studies of the test showed no significant alteration in the performance of the antibody detection ELISA for up to 3 months without the addition of any preservatives. These results provide enough confidence in this antigen to utilize it for the development of a suitable antibody detection-based diagnostic test. We observed some clinical filariasis cases to be positive using this test, which is possible since filarial antibody titres of chronic patients remain high for a long time.²¹

We tried to develop an rWbL1 antigen detection assay, but the antigen detection test failed to show significant efficacy when compared with the IgG4 antibody detection ELISA (data not shown). Therefore, rather than antigen detection, the antibody detection-based assay was shown to be effective.

In conclusion, the present study showed the diagnostic potential of an immunoassay developed using rWbL1 filarial antigen for the detection of active filarial infection, with evidence of reasonably high sensitivity and specificity as well as positive and negative predictive values. Therefore the filarial antibody detection assays developed from this study might be proposed not only as a tool for epidemiological surveillance to monitor the transmission of infection following MDA in filarial elimination programs, but also as a clinically useful diagnostic test. A large multicentric study is further required to validate this test and could prove beneficial for its early utilization in clinics and in transmission monitoring surveys.

Authors' contributions: BVSP, VK and KG designed the study protocol. BVSP and VK performed the experiments. PSY, MSC and DVL contributed to sample collection and the experiments. BVSP, VK and KG drafted the manuscript. All authors critically revised the manuscript and read and approved the final version. BVSP and KG are guarantors of the paper.

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Competing interests: None declared.

Ethical approval: The protocol was approved and registered by the Human Ethics Committee of Mahatma Gandhi Institute of Medical Sciences, Sevagram, India.

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